

Project Title: Scalable Web Application Hosting on AWS

Objective

Design and deploy a secure, highly available, and scalable web application infrastructure for a client launching a nationwide e-commerce platform.

Architecture Components

- **Route 53** – Domain registration and DNS routing
 - **S3** – Hosting static frontend assets (HTML, CSS, JS)
 - **CloudFront** – CDN for global performance and caching
 - **Elastic Load Balancer (ALB)** – Distributes traffic to EC2 instances
 - **EC2 Auto Scaling Group** – Hosts backend application servers
 - **RDS (MySQL/PostgreSQL)** – Managed relational database
 - **VPC** – Custom network with public/private subnets across 2 AZs
 - **Security Groups & NACLs** – Control traffic at instance and subnet level
 - **IAM Roles & Policies** – Secure role-based access
 - **CloudWatch + SNS** – Monitoring and alerts
-

Security Measures

- HTTPS via **ACM** certificates
 - Only ALB exposed to the internet
 - EC2 and RDS in private subnets
 - IAM principle of least privilege
-

Step-by-Step Design: Scalable Web App on AWS

1. Domain & DNS – Route 53

- Register or transfer a domain in Route 53.
 - Create a **hosted zone** for DNS records.
 - Add an **A/alias record** pointing to CloudFront or Load Balancer.
-

2. Static Content – S3 + CloudFront

- Upload your static website files (HTML/CSS/JS) to an **S3 bucket**.
 - Enable **static website hosting** on the bucket.
 - Set up a **CloudFront distribution**:
 - Origin: your S3 bucket
 - Enable HTTPS using an ACM certificate
 - Configure caching and custom error pages
-

3. Networking – VPC & Subnets

- Design a **custom VPC** with:
 - **2 Public Subnets** (for ALB)
 - **2 Private Subnets** (for EC2 & RDS)
 - Add **Internet Gateway** for public subnet access
 - Add **NAT Gateway** for private subnet internet access
-

4. Load Balancer (Application Load Balancer)

- Create an **Application Load Balancer (ALB)**:
 - In public subnets (in two AZs)
 - Target group: backend EC2 Auto Scaling Group
 - Update security group to allow inbound HTTP/HTTPS
 - Register EC2 instances to the ALB
-

5. Backend – EC2 Auto Scaling

- Create a **launch template** for your EC2 app servers:
 - AMI with your app
 - IAM role
 - Security group (allow only ALB inbound)
 - Create an **Auto Scaling Group**:
 - Set desired, min, and max instance counts
 - Configure health checks
 - Enable scaling policies (CPU-based or schedule-based)
-

6. Database – Amazon RDS

- Launch **RDS instance** (e.g., MySQL/PostgreSQL):
 - In **private subnets** across 2 AZs
 - Enable automated backups
 - Use parameter groups for tuning
 - Configure security group to allow traffic **only from EC2 app instances**
-

7. Security – IAM, SGs, NACLs

- Use **Security Groups** to control EC2, RDS, and ALB traffic.
 - Create **IAM roles**:
 - For EC2 to access S3, CloudWatch, etc.
 - Least privilege policy
 - Optional: NACLs for subnet-level traffic control
-

8. Monitoring & Alerts

- Enable **CloudWatch** for:
 - EC2 CPU/memory metrics
 - Application logs
 - Create **CloudWatch Alarms**
 - Set up **SNS** to send email/SMS alerts on high usage, downtime, or failures
-